QPCODE: 25901638

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BCA (Honours) REGULAR EXAMINATION, APRIL 2026
SECOND SEMESTER
Core Course (CC) - MG2CCRBCA102 - OPERATING SYSTEMS
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

SCHEME FOR VALUATION

Part A

Very Short Answer Questions
Answer ***all*** questions
Each question carries **2** *marks*

1. List any two process control system calls [U] / [CO1]

•Create Process
•Terminate Process

OR

Any two from the following:
End, Abort, Load, Execute, Get Process Attributes, Set Process Attributes, Wait For Time, Wait Event, Signal Event, Allocate And Free Memory.

2. Name any two CPU scheduling algorithms that use preemptive scheduling. [An] / [CO2]

Any two of the following:

- Shortest-Job-First Scheduling
- Round Robin Scheduling
- Priority scheduling

3. Explain the different approaches for deadlock detection [U] / [CO3]

• Single Instance of Resources: Use a Wait-For Graph (WFG). A cycle in the graph indicates a deadlock.

• Multiple Instances of Resources: Use a Detection Algorithm similar to the Banker's algorithm. It checks available, allocation, and request matrices to identify deadlocked processes.

4. How can a process execute its critical section? [U] / [CO3]

A process can execute its critical section by following the critical section problem structure, ensuring proper synchronization:
Explanation with Entry Section, Critical Section, Exit Section and Remainder Section

5. How does shares pages help in memory management? [U] / [CO4]

Shared pages help in memory management by allowing multiple processes to share the same physical memory instead of keeping separate copies.

- Reduces memory usage: Common code (e.g., libraries) is stored once and shared.
- Improves efficiency: Saves space and allows faster process creation (e.g., shared libraries)

(2X 5 =10 marks)

Part B

Short Answer Questions

Answer ***any 5*** questions

*Each question carries **6** marks*

6. Discuss dual mode operation in operating system. [U] / [CO1]

Dual mode operation is a mechanism that allows an operating system to protect system resources by running programs in two different modes:

- User Mode: Used for executing application programs. Has restricted access to hardware and system resources. Cannot execute privileged instructions.
- Kernel Mode (Supervisor Mode): Used by the operating system**.**
- Mode Bit: A special hardware bit indicates the current mode (user = 0, Kernel = 1).Ensures the system knows which mode is active.

Explanation with figure give full marks.

7. Write notes on layered operating system structure [K] / [CO1]

A layered operating system structure divides the operating system into a hierarchy of layers, where each layer performs a specific function and interacts only with its adjacent layers.

Layer 0: Hardware
Layer 1: CPU scheduling & memory management
Layer 2: I/O management
Layer 3: File system
Top Layer: User interface / applications

With suitable diagram and explanation of each layer -5 marks.

8. Distinguish a process and job and explain the queue s and schedules related to them. [U] / [CO2]

Difference between job & process – 1 mark
Queues: Job Queue, Ready Queue, Device (I/O) Queue - with explanation -2marks
Schedulers: Long-Term Scheduler (Job Scheduler, Short-Term Scheduler (CPU Scheduler, Medium-Term Scheduler- with explanation -2marks

9. Explain the concept of threads [U] / [CO2]

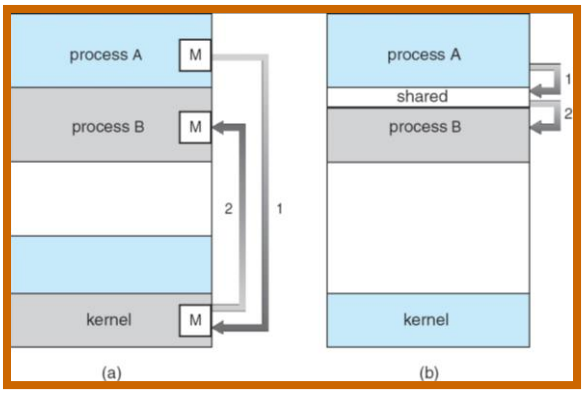
A thread is the smallest unit of execution within a process, often called a lightweight process. A process can have multiple threads, which share the same memory space, code, and resources but have separate registers and stack. 2 marks
Types of Threads: User-level threads and Kernel-level threads – 2 marks
Applications: 1 mark

10. Define a semaphore. Explain its basic operations and use in process synchronization. [U] / [CO3]

- Semaphore definition – 1marks
- Types: Binary semaphore (0/1), Counting semaphore. - with explanation 2 marks
- Operations: wait (P) and signal (V)- with explanation 2 marks

11. Differentiate between shared memory and message passing in IPC. [U] / [CO3]

- **Shared memory** requires communicating processes to establish a region of shared memory .Information is exchanged by reading and writing data in the shared memory
- **Message passing System**: If *P* and *Q* wish to communicate, they need to: establish a *communication link* between them. Exchange messages via send/receive.



Description of each 2 marks + diagram -1 mark

12. Explain the concept of swapping with a diagram. [U] / [CO4]

Swapping is a memory management technique where processes are temporarily moved from main memory (RAM) to secondary storage (swap space) and brought back when needed. Helps in improving memory utilization.

Explanation with swap in, Swap out – 3 marks, Diagram -2 marks

(6X5=30 Marks)

Part C

Essay Questions

*Answer **any 2** questions*

*Each question carries **15** marks*

13. What is the purpose of context switching in process scheduling? [An / CO2]

- Context switching is the process of saving the state of one process and loading the state of another.
  - Enables multitasking
  - Ensures CPU sharing among processes
  - Improves system responsiveness
  - Supports Process Scheduling :Works with scheduling algorithms like Round Robin Scheduling and Priority Scheduling

Context switching definition and steps: 6 marks

Purpose of Context switching in scheduling with example -9 marks.

14. Explain deadlock prevention techniques in detail [U] / [CO3]

Definition of deadlock – 3 marks

Four necessary conditions – 6 marks

Prevention techniques explanation – 6 marks

15. Describe the segmentation technique in memory management. [A] / [CO4]

Segmentation divides memory into logical units called segments such as code, data, and stack. Each segment has a base address and limit stored in a segment table. It allows better memory organization and protection.

Definition of segmentation – 3 marks
Structure (segments, base, limit, segment table) – 5 marks
Working mechanism – 4 marks
Advantages & disadvantages – 3 marks **(15X 2= 30 Marks)**

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